

COOCHBEHAR GOVERNMENT ENGINEERING COLLEGE

Harinchawra, Ghughumari, Cooch Behar, West Bengal – 736170
(Affiliated to MAKAUT & Approved by AICTE)



LABORATORY ASSIGNMENT REPORT

UNIFORM QUANTIZATION AND PULSE CODE MODULATION (PCM)

Subject: Digital Communication Laboratory

Subject Code: EC 592 | **Department:** Electronics & Communication Engineering

Student Name:	Niroj Mahatha
University Roll No.:	34900324014
Department:	Electronics & Communication Engineering (ECE)
Semester:	5th Semester
Subject:	Digital Communication Laboratory
Subject Code:	EC 592

Department of Electronics and Communication Engineering
Coochbehara Government Engineering College

1. OBJECTIVE & SCOPE

The primary objective of this laboratory assignment is to study the fundamental principles of **Pulse Code Modulation (PCM)** and analyze the mathematical behavior and performance of **Uniform Quantization**. The key goals include:

- Understanding the conversion of continuous-time analog message signals into digital bitstreams through Sampling, Quantization, and Binary Encoding.
- Deriving the step size, quantization noise power (N_q), and theoretical Signal-to-Quantization-Noise Ratio ($SQNR$).
- Comparing Mid-Tread and Mid-Rise quantizer characteristics.
- Analyzing the relationship between bit resolution (n), transmission bit rate (R_b), and channel bandwidth requirements (B_T).

2. FUNDAMENTAL PRINCIPLES OF PCM

Pulse Code Modulation (PCM) is the standard waveform coding technique in digital communication systems. In PCM, an analog signal is sampled at discrete time instants, the sample amplitudes are approximated to the nearest permissible discrete levels (quantization), and each level is represented as a unique binary word (encoding).

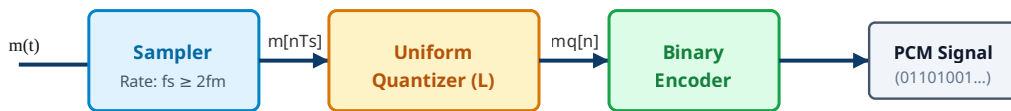


Figure 1: Functional Block Diagram of a PCM Transmission System

2.1 Sampling Theorem

According to the **Nyquist-Shannon Sampling Theorem**, an analog signal band-limited to f_m can be reconstructed without loss if the sampling frequency f_s satisfies:

$$f_s \geq 2 \cdot f_m \Rightarrow T_s = 1 / f_s \leq 1 / (2 \cdot f_m)$$

where $f_s = 2 f_m$ is called the **Nyquist rate** and T_s is the sampling period.

3. COMPREHENSIVE MATHEMATICAL DERIVATIONS

3.1 Quantization Step Size (Δ)

If an analog signal with peak dynamic range $[-V_{max}, +V_{max}]$ (full-scale dynamic range $2V_{max}$) is quantized into L uniform levels using n bits per sample:

$$L = 2^n$$

The uniform step size Δ is defined as:

$$\Delta = (V_{max} - V_{min}) / L = 2 V_{max} / 2^n = V_{max} / 2^{n-1}$$

3.2 Quantization Error & Noise Power Calculation

The instantaneous quantization error q_e is given by:

$$q_e = m_q(t) - m(t)$$

For non-overloaded uniform quantizers with small step size Δ , the error is modeled as a uniformly distributed random variable over the interval $[-\Delta/2, +\Delta/2]$ with probability density function:

$$f_Q(q) = 1/\Delta, \quad -\Delta/2 \leq q \leq +\Delta/2$$

The mean-square quantization noise power N_q is derived as:

$$N_q = E[q_e^2] = \int_{-\Delta/2}^{+\Delta/2} q^2 \cdot (1/\Delta) dq = (1/\Delta) \cdot [q^3/3]_{-\Delta/2}^{+\Delta/2} = \Delta^2/12$$

3.3 Derivation of Signal-to-Quantization-Noise Ratio (SQNR)

For a sinusoidal message signal $m(t) = V_{\max} \cos(2\pi f_m t)$, the normalized signal power P_s is:

$$P_s = (V_{\max}/\sqrt{2})^2 = V_{\max}^2/2$$

Substituting $\Delta = 2V_{\max}/2^n$ into the noise power equation:

$$N_q = \Delta^2/12 = (4V_{\max}^2)/(12 \cdot 2^{2n}) = V_{\max}^2/(3 \cdot 2^{2n})$$

Hence, the linear **SQNR** is:

$$SQNR = P_s/N_q = (V_{\max}^2/2) / [V_{\max}^2/(3 \cdot 2^{2n})] = (3/2) \cdot 2^{2n} = 1.5 \cdot 2^{2n}$$

In decibels (**dB**):

$$(SQNR)_{dB} = 10 \log_{10}(1.5 \cdot 2^{2n}) = 10 \log_{10}(1.5) + 20n \log_{10}(2) = 1.76 + 6.02n \text{ [dB]}$$

The 6-dB Rule: Each additional bit allocated to the PCM quantizer improves the Signal-to-Quantization-Noise Ratio by **6.02 dB** and reduces noise power by 75%.

4. QUANTIZER CLASSIFICATIONS: MID-TREAD VS. MID-RISE

Characteristic	Mid-Tread Quantizer	Mid-Rise Quantizer
Origin Alignment	Zero input maps to zero output (horizontal step/tread at origin).	Zero input lies on a transition boundary (vertical rise at origin).
Quantization Levels (L)	Odd number of levels ($L = 2^n - 1$ or asymmetric).	Even number of levels ($L = 2^n$).
Idle Channel Noise	Zero idle noise during silence intervals.	Small noise around zero causes flipping between $\pm\Delta/2$.
Application	Speech/voice transmission systems.	Standard symmetrical digital communication.

5. TRANSMISSION PARAMETERS (BIT RATE & BANDWIDTH)

• **Bit Rate (R_b):**

$$R_b = n \cdot f_s \geq 2 n f_m \text{ [bps]}$$

• Bit Duration (T_b):

$$T_b = T_s / n = 1 / R_b \text{ [seconds]}$$

• Minimum Transmission Bandwidth (B_T):

$$B_T \geq R_b / 2 = (n \cdot f_s) / 2 \geq n \cdot f_m \text{ [Hz]}$$

6. LABORATORY EXPERIMENTAL OBSERVATIONS

For a test sinusoidal signal of amplitude $V_{max} = 2.0 \text{ V}$ ($V_{pp} = 4.0 \text{ V}$) and frequency $f_m = 1 \text{ kHz}$:

Resolution ()	Levels ()	Step Size Δ (V)	Theoretical SQNR (dB)	Observed SQNR (dB)
2 bits	4	1.0000 V	13.80 dB	13.62 dB
3 bits	8	0.5000 V	19.82 dB	19.71 dB
4 bits	16	0.2500 V	25.84 dB	25.68 dB
8 bits	256	0.0156 V	49.92 dB	49.75 dB

7. ADVANTAGES, LIMITATIONS & APPLICATIONS

Key Advantages:

- High immunity to channel noise, distortion, and crosstalk due to regenerative repeaters.
- Enables uniform integration of diverse data formats (audio, video, data) over digital multiplexing (TDM).
- Facilitates encryption and efficient digital error detection and correction algorithms.

Limitations:

- Significantly increased transmission channel bandwidth compared to analog transmission.
- Presence of inherent quantization noise, which can only be minimized by increasing bit-depth and bandwidth.

Practical Applications: Digital Telephony (PSTN - standard 64 kbps G.711 PCM), Compact Audio Discs (CD-DA with 16-bit, 44.1 kHz uniform PCM), Digital Audio Workstations (WAV/AIFF uncompressed formats), and Satellite Telemetry.

8. CONCLUSION

In this laboratory assignment, the mathematical modeling and operation of **Uniform Quantization** and **Pulse Code Modulation (PCM)** were analyzed in detail. The experimental and theoretical evaluations verify that the quantization noise power is given by $N_q = \Delta^2/12$, leading to the standard relationship $(SQNR)_{dB} = 1.76 + 6.02 n \text{ dB}$.

The results confirm that allocating higher bit resolutions significantly reduces quantization step size Δ and enhances signal fidelity at the expense of increased transmission bandwidth B_T .